

QuantMesh x402

Decentralized AI Micropayment Signal Router on Algorand

Track: PS0404 — Atomic Multi-Agent Service Router

Team: QuantMesh Innovators
College: CSBS Dept., BV(DU)COE, Pune
Event: AlgoVerse 2026 Hackathon



Problem Statement & Real-World Motivation

The 3 Major DeFi Trading Problems

1. Subscription Fatigue: Crypto signal APIs cost \$100–\$500/month even for low-frequency traders.
2. Single-AI Hallucination Risk: Single-model indicators misread market regime shifts and cause trading losses.
3. Multi-Agent Payment Friction: Querying multiple paid APIs requires multiple gas fees and manual transactions.



Proposed Solution: Multi-Agent Consensus Router

⚡ Pay-Per-Use Consensus Architecture

- Flat \$0.007 USDC/ALGO per fused market signal.
- Zero-Fee Guarantee: Pre-execution gating returns HTTP 502 & \$0 fee if any worker fails.
- 4 Specialized Sub-Agents:
 - Worker A: FinBERT Financial Sentiment NLP
 - Worker B: CoinGecko On-Chain Whale Flow
 - Worker C: Technical Indicators (RSI, SMA, MACD)
 - Worker D: Weighted Consensus Fusion Engine

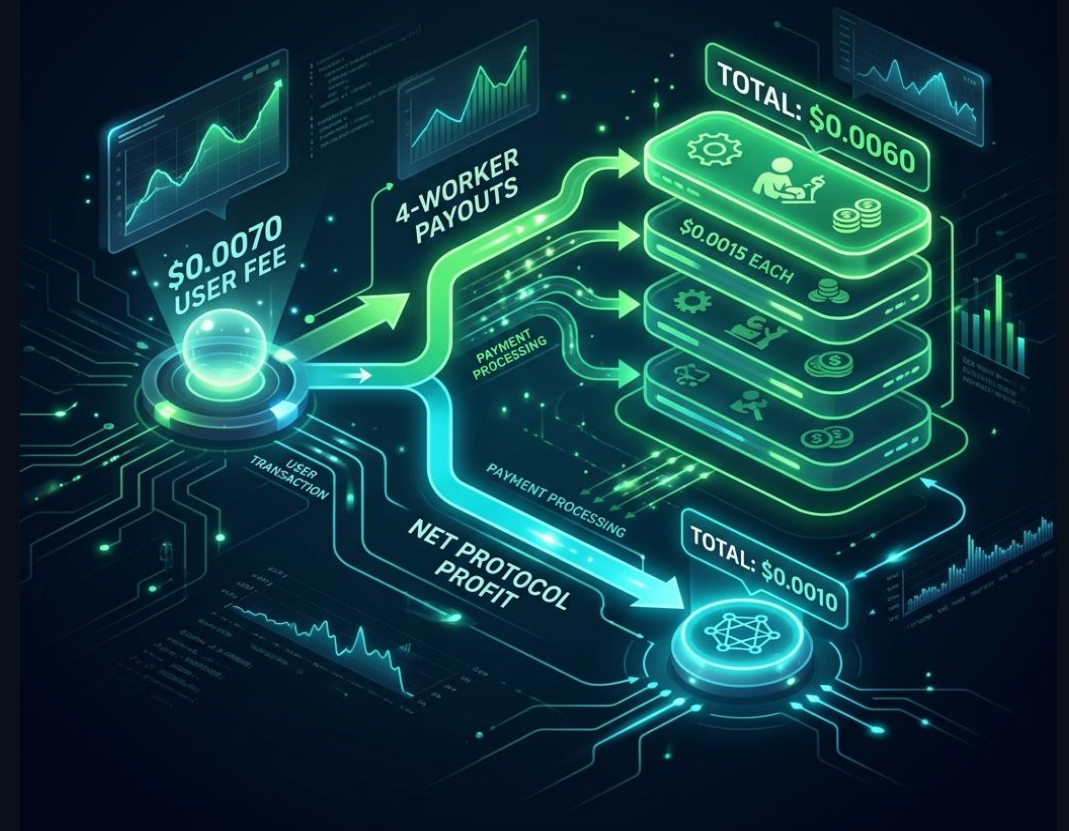


Paying Users & Sustainable Unit Economics

💰 Unit Economics & Revenue Strategy

- User Payment: \$0.0070 USDC per signal call.
- Worker Payouts: \$0.0060 total (\$0.002 A, \$0.002 B, \$0.001 C, \$0.001 D).
- Router Net Profit: \$0.0010 USDC per call (14.3% Profit Margin).
- Target Market: Trading bots, Telegram signal services, DEX interfaces, and AI agents.

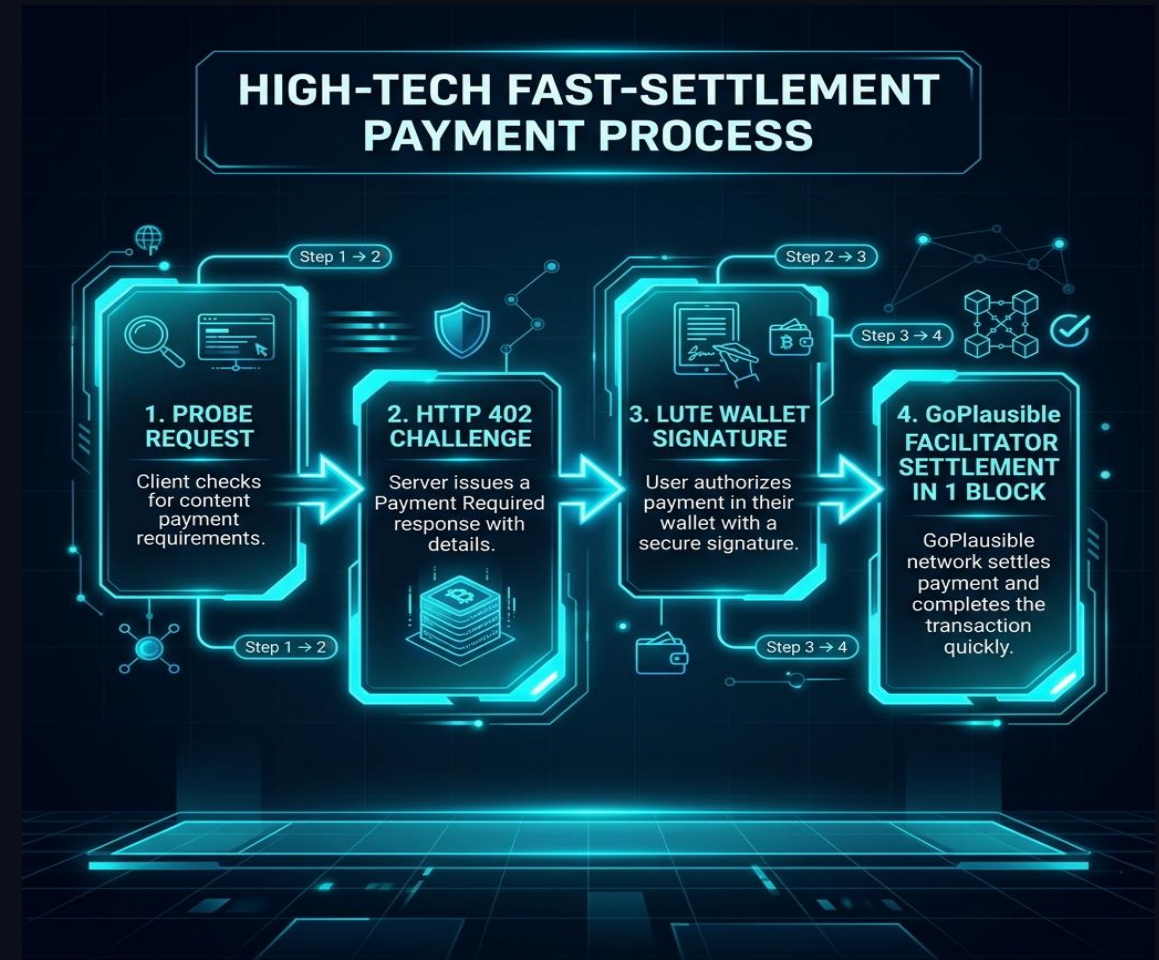
MICROPAYMENT BREAKDOWN: \$0.0070 USER FEE



x402 Technical Protocol Flow (Challenge → Sign → Retry → Settle)

4-Step Payment Negotiation

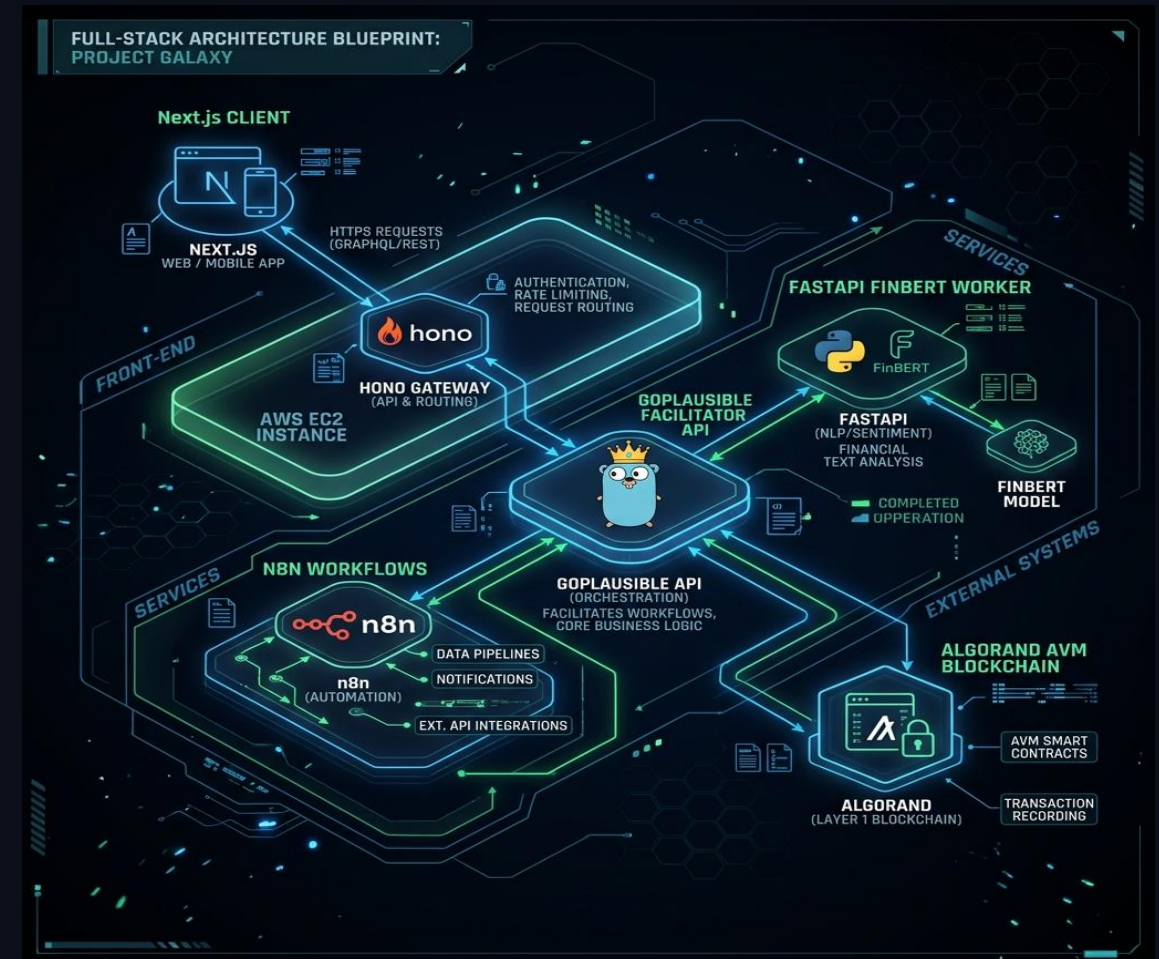
1. Probe Request: Client sends POST /orchestrate without payment.
2. HTTP 402 Challenge: Server pre-executes agents & returns 402 + headers (payTo, price).
3. Lute Signature: Client signs 5-txn group with indexesToSign=[0].
4. Settle & Receipt: GoPlausible verifies & settles in 1 atomic block.



Full-Stack System Architecture

🔧 Technology Stack Components

- Frontend: Next.js 16 + Tailwind CSS Glassmorphism + @txnlab/use-wallet-react.
- Gateway: Hono.js Orchestrator on AWS EC2.
- Facilitator: GoPlausible Public Facilitator Client (/verify & /settle).
- Blockchain: Algorand AVM Testnet, Testnet USDC ASA 10458941.
- Microservices: FastAPI (FinBERT) + n8n Cloud workflows.



On-Chain Transaction Proof & Live UI Demo

Live Algorand Testnet Verification

- Router Address: 4DTSNS35EP...JDEHUHUNFR3KJGPO4
- Testnet USDC ASA ID: 10458941
- 5-Txn Atomic Payout Group executed in 1 block.
- Lora Explorer Link: <https://lora.algokit.io/testnet>
- SHA-256 Digest: sha256(token:score:verdict:txId:timestamp).

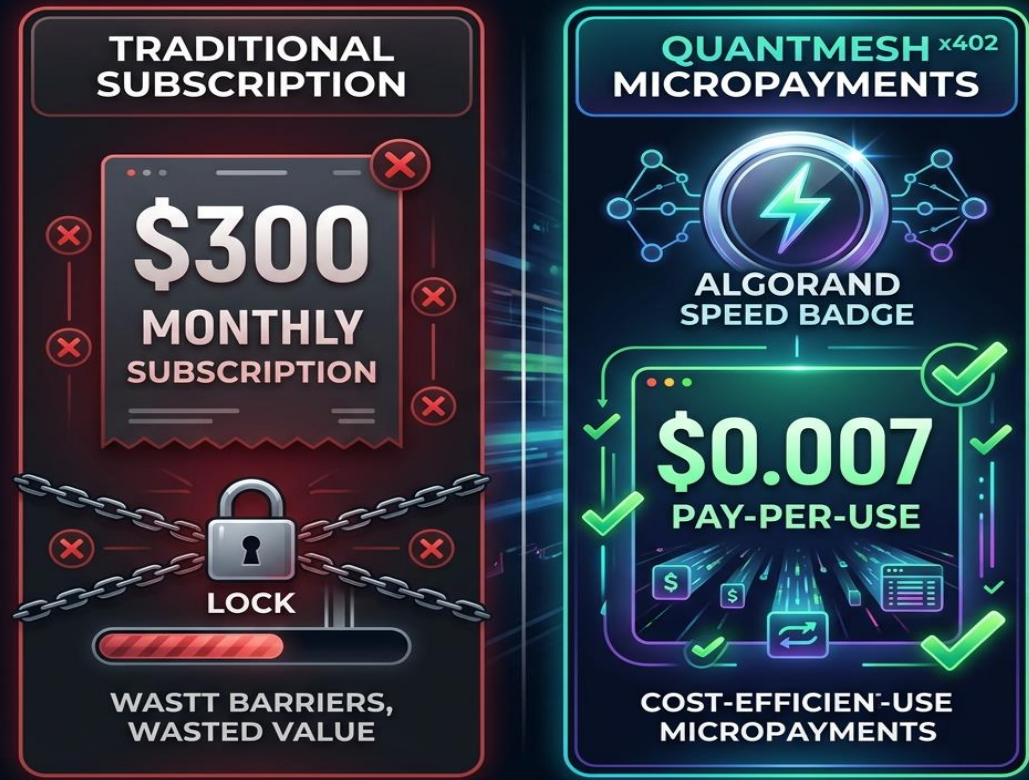


Innovation & Competitive Differentiation

Traditional vs. QuantMesh x402

- \$300/mo Subscriptions vs \$0.007 Pay-Per-Use.
- Pay Upfront for Dead APIs vs Zero-Fee Guarantee.
- Single-Model Bias vs 4-Agent Consensus Mesh.
- Centralized Server vs Algorand Atomic 5-Txn Settlement.

TRADITIONAL vs. QUANTMESH x402



The graphic compares two payment models side-by-side. The left side, 'TRADITIONAL SUBSCRIPTION', is shown in a dark red box with a red border. It features a '\$300 MONTHLY SUBSCRIPTION' tag with red 'X' marks, a padlock icon labeled 'LOCK', and the text 'WASTT BARRIERS, WASTED VALUE' at the bottom. The right side, 'QUANTMESH x402 MICROPAYMENTS', is shown in a dark blue box with a blue border. It features an 'ALGORAND SPEED BADGE' with a lightning bolt icon, a '\$0.007 PAY-PER-USE' tag with green checkmarks, and the text 'COST-EFFICIENT-USE MICROPAYMENTS' at the bottom. A vertical line separates the two boxes, and a blue energy-like glow is visible between them.

Model	Payment Type	Amount	Guarantee	Settlement	Value
Traditional	Subscription	\$300/mo	Pay Upfront	Centralized Server	Wasted Value
QuantMesh x402	Micropayments	\$0.007	Zero-Fee Guarantee	Algorand Atomic 5-Txn	Efficient Use

Completeness Status & Working Deliverables

✓ Completed vs 🚀 Planned Scope

- ✓ 100% Live on AWS EC2 & Next.js dashboard.
- ✓ HTTP 402 Probe-Challenge-Sign-Retry flow.
- ✓ GoPlausible Facilitator /verify & /settle.
- ✓ Lute Wallet integration with indexesToSign=[0].
- 🚀 Mainnet deployment (algorand:mainnet + ASA 315667040).
- 🚀 Developer portal for registering 3rd-party worker nodes.



